ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Drawings: award marks only for components that are drawn with the correct symbol and connected correctly. Accept other correct answers.

QUESTION 1

1.1 A diode is a semiconductor device ✓ that allows current to flow in one direction only. ✓ (2)

1.2 • Rectification (changing AC to DC) ✓ (2)
• Protection against reverse polarity / free-wheeling diode across a relay coil ✓
(Any TWO; also accept voltage regulation with a Zener diode, signal or indicator LEDs.)

1.3 An analogue signal changes continuously and can have any value within a range. ✓✓ (4)
A digital signal has only two levels (ON/OFF, 1/0). ✓✓ (2 × 2)

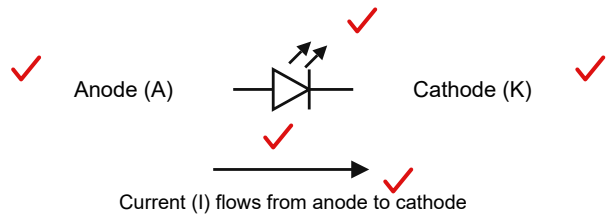
1.4.1 12 V ✓ (1)

1.4.2 2 × 12 = 24 V ✓ (1)

1.4.3 T = 4 ms ✓✓ (2)

1.4.4 $f = 1 \div T \checkmark = 1 \div 0{,}004 \checkmark = 250\ Hz \checkmark$ (3)

1.5



(1 mark each: triangle and bar, light arrows, anode, cathode, current direction) (5)

[20]

QUESTION 2

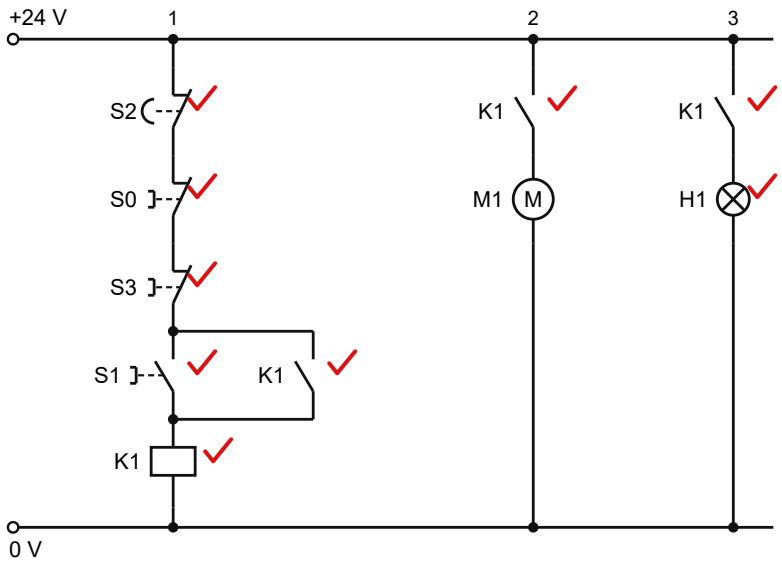
2.1 A stepper motor is a DC motor that turns in fixed steps (angles) ✓ for each pulse it receives from its driver. ✓ (2)

2.2 • Varying the armature voltage ✓ (3)
• Pulse-width modulation (PWM) ✓
• Varying the field current / field resistance ✓
(Any THREE; also accept a series resistor in the armature circuit.)

2.3 A limit switch detects the position of a moving part ✓ and switches the circuit on or off when the part reaches that position (e.g. the end of a stroke). ✓ (2)

2.4 • A small control current can switch a large load current. ✓ (3)
• The control circuit is electrically isolated from the motor circuit. ✓
• One relay can switch several circuits at the same time. ✓
(Any THREE; also accept: low-voltage control for safety, remote switching.)

2.5



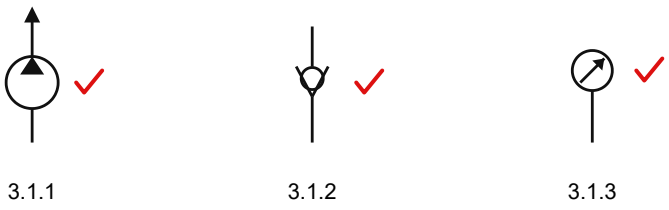
(9 × 1 for the components shown with ticks, 1 for correct series and parallel connections)

(10)

[20]

QUESTION 3

3.1

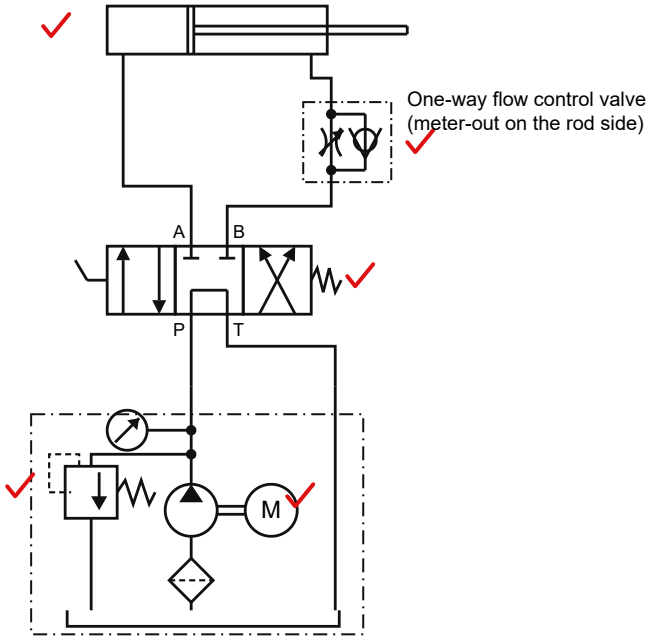


(2 marks per symbol: 1 for the shape, 1 for the correct detail)

(6) |

3.2 3.2.1 Double-acting cylinder ✓ (6)
3.2.2 4/3-way directional control valve (lever operated, tandem centre) ✓
3.2.3 Pressure gauge ✓
3.2.4 Pressure relief valve ✓
3.2.5 Electric motor ✓
3.2.6 Suction filter ✓ (6 × 1)

3.3



(Cylinder 1, 4/3-way tandem-centre valve 2, one-way flow control valve on the rod side 2, power pack 2, correct connections 1)

(8)

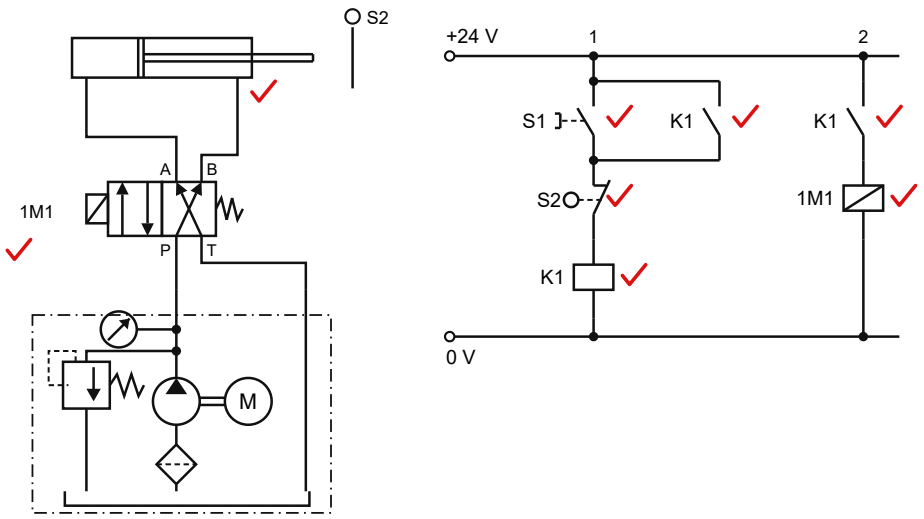
[20]

QUESTION 4

4.1 When the coil is energised, the solenoid becomes an electromagnet ✓ that moves the valve spool and changes the direction of the oil flow. ✓ (2)

4.2 A normally open (NO) push button is open when it is not pressed and closes (passes current) when it is pressed. ✓✓ (4)
A normally closed (NC) push button is closed when it is not pressed and opens (stops current) when it is pressed. ✓✓ (2 × 2)

4.3



(Hydraulic circuit: cylinder 1, 4/2-way solenoid valve with spring return 2, power pack 1. Control circuit: S1 with K1 holding contact in parallel 3, S2 (NC) in series 2, K1 coil 1, K1 contact with 1M1 2. Marker to use discretion; components must be connected correctly.)

(12)

4.4 Interlocking prevents two operations from happening at the same time ✓ (e.g. a forward and a reverse contactor), by placing an NC contact of one in the circuit of the other. ✓ (2)

[20]

QUESTION 5

5.1 Preventive maintenance is planned, regular maintenance ✓ that is done to prevent breakdowns and to extend the life of the equipment. ✓ (2)

5.2
- Switch off and isolate the supply (lock out). ✓
- Disconnect at least one wire of the coil. ✓
- Set the multimeter to the resistance (Ω) or continuity range. ✓
- Connect the probes across the coil terminals. ✓
- A low resistance reading shows the coil is continuous; an infinite (OL) reading shows an open coil. ✓

(5)

5.3.1 Replace the piston seals ✓ and test the cylinder for leaks before use. ✓ (2)

5.3.2 Repair or replace the directional control valve (spool) ✓ and check the system pressure. ✓ (2)

5.3.3 Bleed the air from the system ✓ and check the suction line and oil level so that no new air is drawn in. ✓ (2)

5.4 • Switch off and isolate the power pack (lock out and tag out). ✓ (7)
• Release the pressure in the system. ✓
• Drain the old oil from the reservoir while it is still warm. ✓
• Clean the reservoir and replace or clean the filters. ✓
• Fill with new oil of the correct grade to the correct level. ✓
• Start the system, bleed the air and check for leaks. ✓
• Check the oil level again and top up if necessary. ✓

[20]

TOTAL: 100

END OF MARKING GUIDELINE